

Wi-Fi Controlled Mecanum Bot

Individual Project Documentation

Project Contributor: Inchara M.

Project Context: Hands-on Build Club Robotics Workshop

1. Project Overview

This project involved building a Mecanum wheel robot controlled by an ESP32-C6. The objective is to develop a robot capable of multidirectional movement, including forward, backward, left, right, diagonal movement and rotation, using a custom web-based control interface hosted by the ESP32-C6.

2. Project Objective

The workshop objective is to develop a Mecanum wheel robot car that can move in all specified directions and be controlled through a joystick-style web interface using the ESP32-C6.

3. Hardware Used

Component	Quantity	Purpose
ESP32-C6 Dev Board	1	Main controller and Wi-Fi communication
L298N Motor Driver	2	Controls the DC motors
12V DC Motors (300 RPM)	4	Drives the four wheels
Mecanum Wheels	4	Enables multidirectional movement
Robot Chassis	1	Mechanical platform
12V Battery	1	Motor power supply
Jumper Wires	As needed	Electrical connections
USB Type-C Cable	1	Programming the ESP32-C6

4. Working Principle

The ESP32-C6 is a Wi-Fi-capable microcontroller and hosts a web page containing a virtual joystick/control interface. A user selects a direction in the browser, and the interface sends a command to the ESP32-C6. The ESP32 interprets the command and sets the motor control pins to produce the required wheel movement.

The Mecanum wheels use angled rollers. Coordinating the rotation of the four wheels allows the robot to move in multiple directions without first turning the chassis.

5. Supported Movements

- Forward
- Backward
- Left
- Right
- Forward-Left
- Forward-Right
- Backward-Left

- Backward-Right
- Rotate-Left
- Rotate-Right
- Stop

6. Software Setup

- Arduino IDE is used for programming the ESP32-C6.
- The ESP32 board package is installed through Arduino IDE's Board Manager.
- The ESP32-C6 Dev Module/Dev Board is selected for programming.
- The program contains the Wi-Fi credentials and web-server functionality.
- After uploading, the Serial Monitor is used to view the Wi-Fi connection and IP address.

7. Web Control Interface

The supplied project interface provides directional controls for the four cardinal directions, four diagonal directions, rotation, and stopping. It also includes a speed slider with PWM values from 0 to 255. The browser sends movement and speed requests to the ESP32 web server.

8. Motor-Control Implementation

The project uses two L298N motor-driver modules for the four motors. The ESP32-C6 supplies the control signals, while the motor drivers handle the motor-control interface. The supplied program assigns separate direction and enable/PWM pins for the four motors.

Motor	IN1	IN2	Enable/PWM
M1	GPIO 15	GPIO 23	GPIO 8
M2	GPIO 6	GPIO 7	GPIO 9
M3	GPIO 4	GPIO 5	GPIO 10
M4	GPIO 21	GPIO 22	GPIO 11

Note: The table above records the pin assignments present in the workshop's supplied program. Actual physical wiring should be checked against the workshop circuit diagram and the assembled board.

9. Build and Testing Process

- Mount the motors in their respective chassis slots.
- Secure the motors using the provided hexagonal nuts.
- Fit the brass adapters into the Mecanum wheel hubs.
- Secure the wheels and adapters using the provided bolts and grub screws.
- Repeat the assembly for all four motors and wheels.
- Connect the four motors to the motor-driver modules.
- Connect the motor-driver control inputs to the ESP32-C6 according to the provided pinout.
- Connect the motor power supply to the drivers and the ESP32-C6 to the computer through USB for programming.
- Upload the program and check the assigned IP address through Serial Monitor.
- Open the ESP32-C6 IP address in a browser on the same Wi-Fi network and test the movement controls.

10. My Project Contribution

My work on the project involved participating in the setup and integration of the ESP32-C6, Mecanum wheels, DC motors and L298N motor drivers; setting up the Arduino environment and required ESP32 software; and understanding the web-based control and motor-movement logic used by the robot.

11. Key Learning

- ESP32-C6 setup and programming using Arduino IDE
- Basic DC motor and L298N motor-driver interfacing
- Mecanum wheel movement and coordinated motor control
- Wi-Fi communication between a microcontroller and browser
- Web-server-based robotic control
- Integration of mechanical, electronic and software components

12. Future Scope

The workshop's post-project activity includes implementing precise-angle movement, such as 15° or 25°, and identifying an interesting college use case for the Mecanum robot. Further development can build on these activities.

13. Project Status

Hardware assembled and initial software environment configured as part of the hands-on robotics workshop. The project is suitable for further testing, movement refinement and additional features.

Source: Build Club Workshop Manual — Mecanum Wheeled Robot. This documentation has been prepared as an individual project record based on the supplied workshop manual and the project details provided by the contributor.